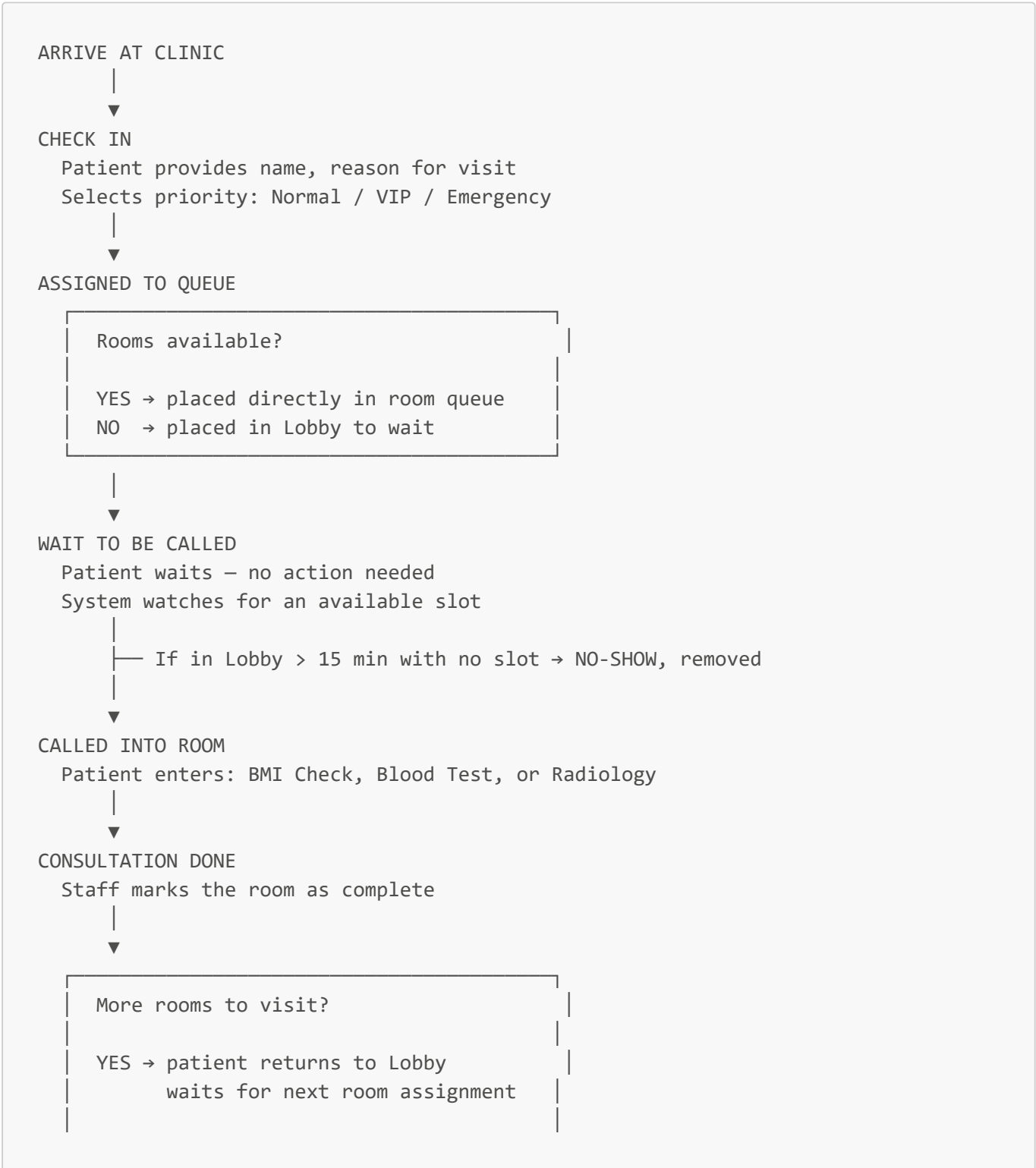


Wellness Center — Business Flow

Overview

The system has three actors: the **Patient**, the **Staff / System**, and the **AI (Gemini)**. The AI is only involved in one moment — when the system detects a queue imbalance and needs to decide which patient to move.

1. Patient Journey



NO → visit complete, patient exits

At any point during waiting, the patient **may be transferred** to a less busy room.
This is invisible to them — they simply receive a notification on their phone.

2. Where AI Is Applied

The AI is involved in **one specific moment** only:

QUEUE IMBALANCE DETECTED

Triggered by:

- Staff marks a room Done
- Emergency patient checks in
- Automatic scan every 60 seconds
- Manual trigger

SYSTEM CHECKS LOAD GAP

Compare queue load between rooms of the same type
(e.g. Blood Test Room 1 vs Blood Test Room 3)

Gap too small → do nothing

FILTER ELIGIBLE PATIENTS

Only Normal priority patients
Only those currently waiting (not being seen)
Not the next-in-line (too disruptive)
Not already moved today

0 candidates → do nothing

1 candidate → pick directly, no AI needed

AI CALLED (Gemini)

AI RETURNS A DECISION

APPLY OR SUGGEST

- Suggest mode → staff sees a card, clicks Accept or Decline
- Auto mode → patient is moved immediately

3. AI Input / Output

Input — what Gemini receives

Field	Description
Candidate list	Each eligible patient: name, how long they have waited, their position in the queue
Busy room	Name, current load (minutes of work queued)
Free room	Name, current load
Load gap	Difference between the two rooms
Threshold	Minimum gap required to justify a move

Example prompt sent to Gemini:

```
Busy room:  Blood Test Room 1 – load 48 min
Free room:  Blood Test Room 3 – load 12 min
Gap:        36 min  (threshold: 20 min)
```

Candidates to consider:

- Hoang Van E – waited 31 min – queue position 2
- Le Van F – waited 24 min – queue position 3
- Ngo Van K – waited 19 min – queue position 4

Who should be moved? Pick the fairest choice.
Explain in one sentence.

Output — what Gemini returns

Field	Description
Selected patient ID	Which patient to move
Reason	One plain-language sentence explaining the decision
aiUsed	true — confirms AI made the call

Example response from Gemini:

```
Selected:  Hoang Van E
Reason:    "Hoang Van E has waited the longest and is not
           next in line – moving him causes the least
           disruption and is the fairest choice."
```

Fallback — when AI is unavailable

If Gemini times out or returns an error, the system falls back automatically:

Field	Value
Selected patient	Longest-waiting candidate
Reason	"Auto-selected longest-waiting patient"
aiUsed	false

The system never stops working because of an AI failure.

4. End-to-End Summary

Phase	Patient does	Staff / System does	AI does
Arrival	Check in, select priority	Register, assign to queue	—
Waiting	Wait	Monitor queues, fill empty rooms	—
Consultation	Receive care	Mark room Done, call next patient	—
Rebalancing	Receive notification, Accept or Decline	Detect imbalance, apply or suggest move	Rank candidates, explain decision
Exit	Leave clinic	Remove from system	—